Results outline

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Research questions

1. Does tree growth increase with longer growing seasons?
2. Did growth change with earlier/later start and end of season?
3. Did growth change across species and populations?

Results

1. Summary basic information
   (a) Wildchrokie includes four deciduous tree species from four sites with 227 complete growing seasons (GS)
   (b) GSL ranged from 90 days (*B. alleghaniensis* in 2018) to 173 days (*A. incana* in 2018)
   (c) Phenology data span three years from 2018 to 2020 with 239 leafout and 230 budset observations. The earliest leafout day of year was 112 (2019) and the latest, 157 (2019). The earliest budset day of year was 225 (2018) and the latest, 297 (2018)
   (d) GS thermal sum ranged from 1465.1 GDD (*B. alleghaniensis* in 2018) to 2513.6 GDD, (*A. incana* in 2018)
2. Did longer growing seasons increase growth?
   (a) Longer growing seasons increased growth for 3 species, depending on the metric
   (b) Longer thermal season increased growth for *B. papyrifera* (β = 0.32, UI_{90} [0.04, 0.6]) (Figure 1a (Table S1).
   (c) But longer calendar season increased growth for two additional species (*A. incana*: β = 0.09, UI_{90}[0.04, 0.14], *B. alleghaniensis*: β = 0.07, UI_{90}[0, 0.13] and *B. papyrifera*: β = 0.19, UI_{90}[0.05, 0.33]) (Figure 1b (Table S1).
   (d) Calendar season had a bigger effect on growth than the thermal season (Figure S1).
   (e) *B. alleghaniensis*trended in the same direction; only *B. populifolia*showed clear no effect.
3. Did growth change with earlier/later start and end of season?
   (a) A later start of season decreased growth for two species (*A. incana*: β = -0.18, UI_{90} [-0.23, -0.13] and *B. populifolia*: β = -0.13, UI_{90}[-0.23, -0.02]) (Figure 1c (Table S1).
   (b) A later end of season decreased growth only for *A. incana* (β = -0.09, UI_{90} [-0.15, -0.04]) (Figure 1d (Table S1).
4. Did growth change across species and populations?
   (a) Annual ring widths across years varied from 0.39 mm (*B. alleghaniensis* in 2020) to 13.55 mm (*B. populifolia* in 2018).
   (b) *B. papyrifera* grew the least (β = -3.31, UI_{90} [-7.89, 1.24]) and *B. populifolia* grew the most (β = 1.05, UI_{90} [-3.04, 5.01]) (Figure 3).
   (c) Growth did not change across any site (Figure 2) (Table S2).
5. Did growth change across years, and how did it relate to climate?

(a) Growth was highest on average in 2018, but no cue dominated (it was neither: the wettest; the driest; the shortest; the latest SOS; the earliest EOS nor the highest thermal sum)

(b) Growth was lowest on average in 2020 and that year was: the driest; the shortest; the latest SOS and EOS and had the highest thermal sum

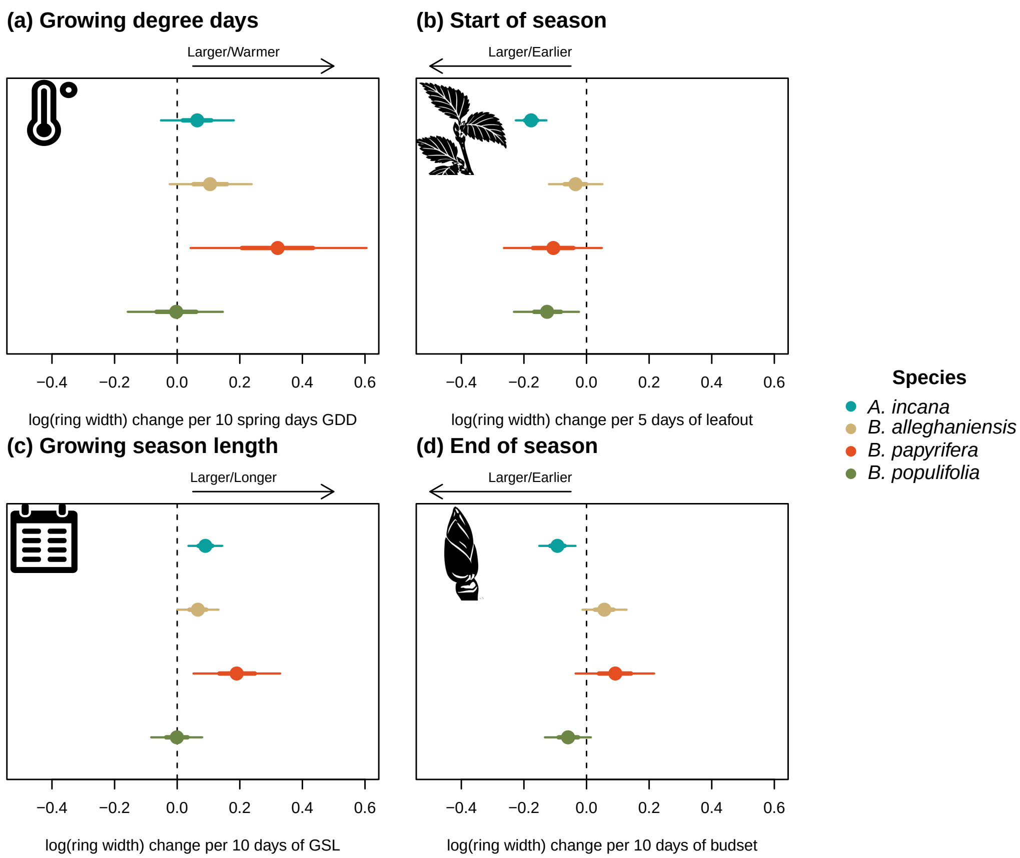

Figure 1: Model slope parameters across the different model predictors for Wildchrokie species. GDD represent the accumulated growing degree days between leafout and budset (a) the models estimates shows log ring width change per 10 average spring days GDD (174.05). GSL is the number of days between leafout and budset (b). SOS represent the start of season, which is leafout (c) and EOS, the end of season through budset (d). All predictors were fit using the same model and number of observations, with only the independent variable changing. The dots represent the mean of the posterior distribution, the thick lines the 50% uncertainty intervals (UI) and the thinner lines, the 90% UI.

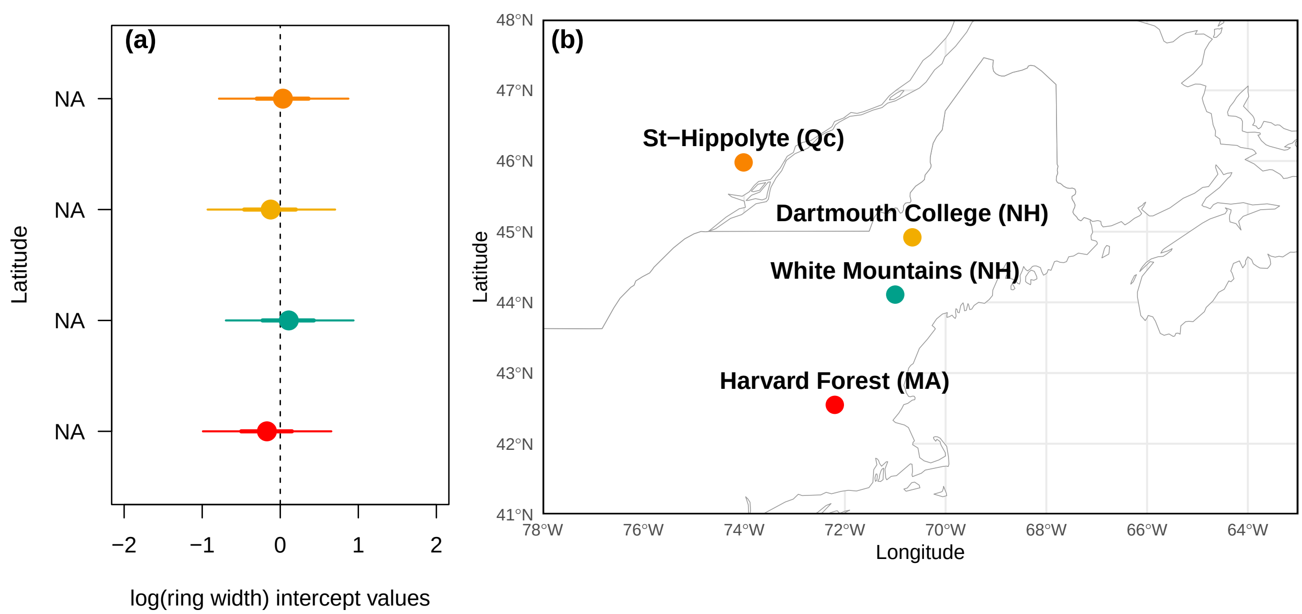

Figure 2: The effect of sites on growth across a latitudinal gradient. (a) Model intercept parameters for each site with growing degree days (GDD) as the predictor. The dots represent the mean of the posterior distribution, the thick lines the 50% uncertainty intervals (UI) and the thinner lines, the 90% UI. The sites are color-coded from (b), th map showing their locations.

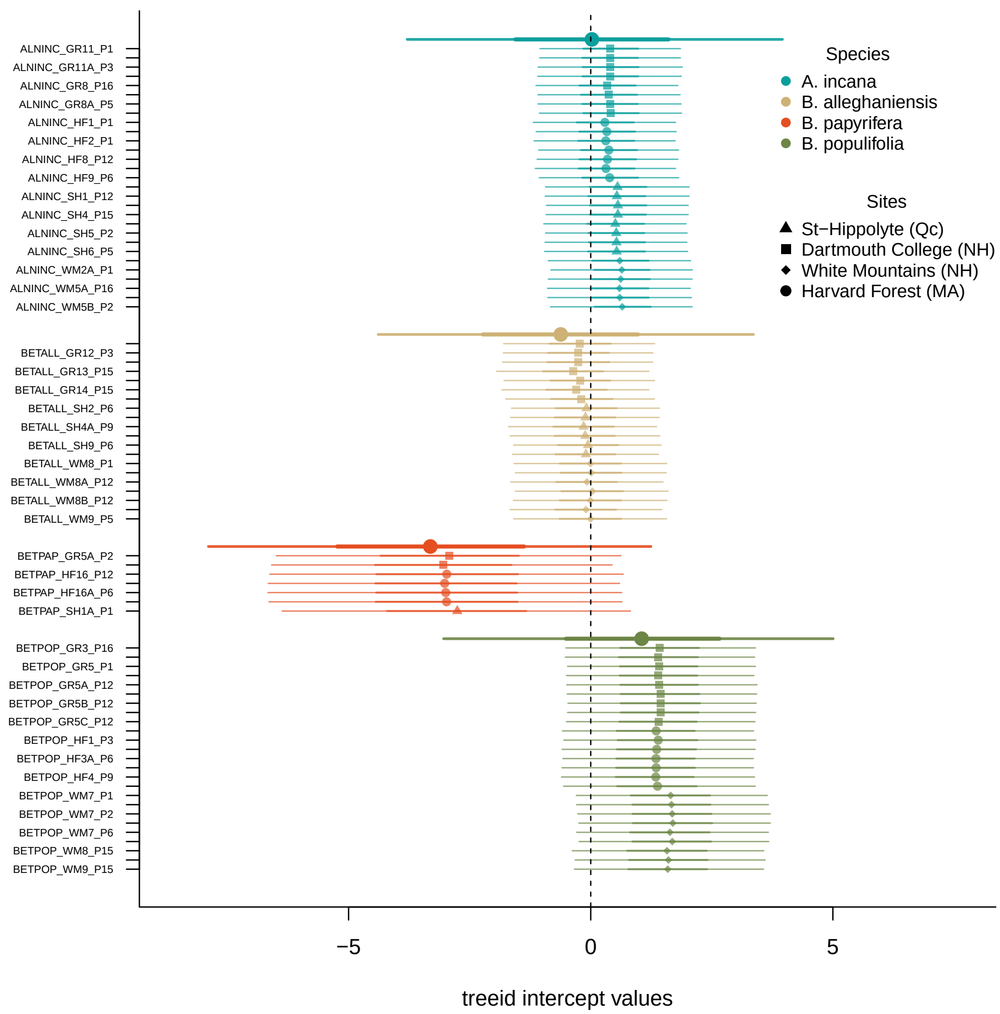

Figure 3: atreeid

Supplemental results

Table S1: Species level slope parameters (β and 90% uncertainty intervals, in square brackets) across the four different predictors for Wildchrokie. GDD: growing degree days, GSL: growing season length, SOS: start of season and EOS: end of season.

Species	GDD	GSL	SOS	EOS
A. incana	β =0.06, [-0.05, 0.18]	β =0.09, [0.04, 0.14]	β =-0.18, [-0.23, -0.13]	β =-0.09, [-0.15, -0.04]
B. alleghaniensis	β =0.10, [-0.02, 0.24]	β =0.07, [0.00, 0.13]	β =-0.04, [-0.12, 0.05]	β =0.06, [-0.01, 0.13]
B. papyrifera	β =0.32, [0.04, 0.60]	β =0.19, [0.05, 0.33]	β =-0.11, [-0.26, 0.05]	β =0.09, [-0.04, 0.22]
B. populifolia	β =-0.00, [-0.16, 0.15]	β =-0.00, [-0.08, 0.08]	β =-0.13, [-0.23, -0.02]	β =-0.06, [-0.13, 0.01]

Table S2: Site level intercept parameters (α and 90% uncertainty intervals, in square brackets) for the GDD model.

Site	Intercept.estimate	Latitude	Longitude
	α =-0.12, [-0.93, 0.7]	44.79	-71.15
	α =-0.17, [-0.99, 0.65]	42.53	-72.19
	α =0.03, [-0.78, 0.87]	45.93	-74.03
	α =0.11, [-0.7, 0.94]	44.11	-71.23

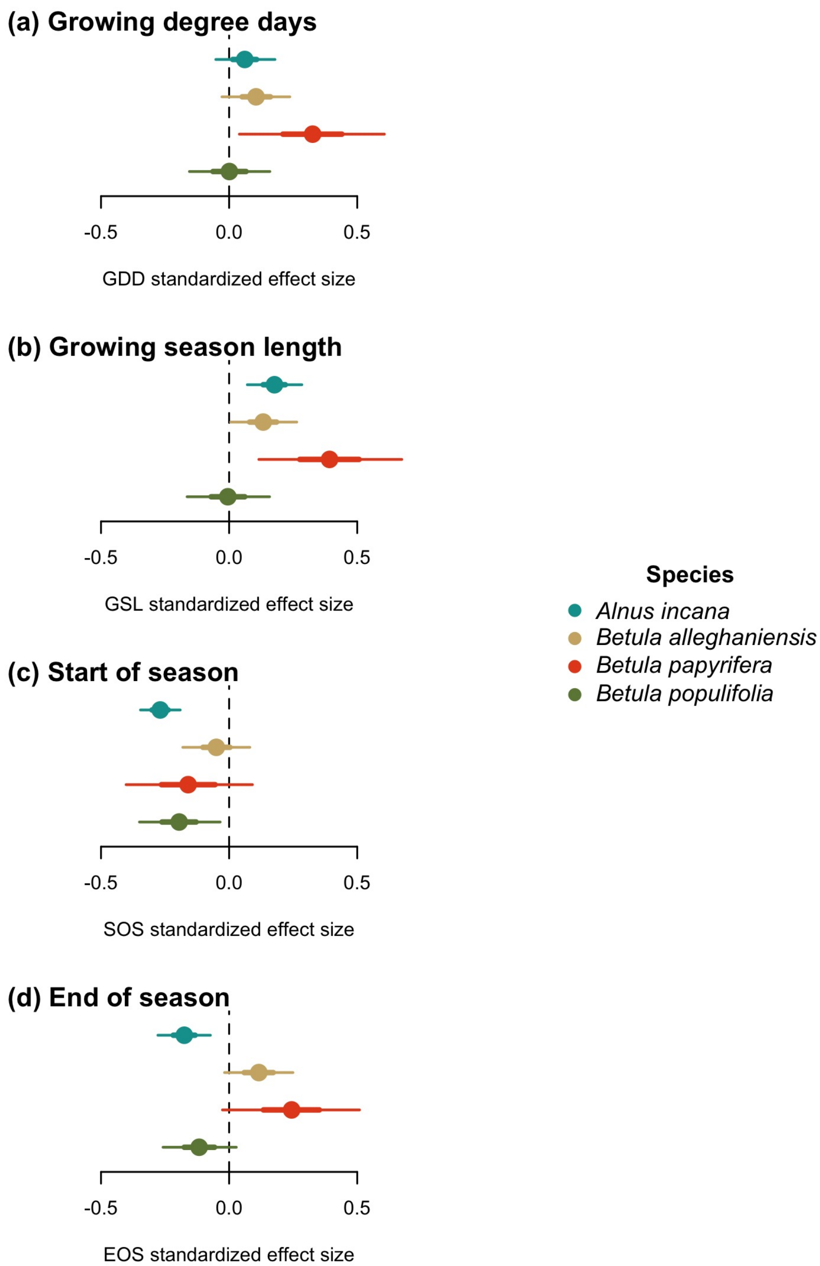

Figure S1: Standardized (z-scored) slope estimates for Wildchrokie. The dots represent the mean of the posterior distribution, the thick lines the 50% uncertainty intervals (UI) and the thinner lines, the 90% UI.